

INTELLIGENT DOCUMENT PROCESSING

Technical Evaluation Report — Docling & Multimodal Extraction

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This report presents a comprehensive technical assessment of Docling's document understanding capabilities across diverse content modalities including rich prose, multi-level tables with spanning cells, embedded charts, architectural diagrams, and mixed-layout pages. The evaluation covers layout analysis accuracy, table structure reconstruction fidelity, image region detection, and end-to-end pipeline throughput. Results are benchmarked against three baseline systems across five document categories.

1. Executive Summary

Modern enterprises process millions of documents annually — contracts, research papers, financial filings, and technical manuals. Extracting structured information from these documents with high fidelity remains a significant engineering challenge. Unlike simple text extraction, production document intelligence requires simultaneous handling of paragraphs, multi-column layouts, nested tables, embedded figures, captions, footnotes, and cross-references.

Docling, developed by IBM Research, addresses this challenge through a two-stage deep-learning pipeline. The first stage performs layout analysis — classifying each page region as text, table, figure, heading, list, or formula. The second stage applies specialized sub-models per region type: TableFormer for table grid reconstruction and an optional VLM captioner for figure understanding. This architecture delivers state-of-the-art results on the DocLayNet and PubTables-1M benchmarks.

1.1 Key Findings

- Table structure reconstruction achieves 91.3% GriTS (Grid Tree Structure) score on PubTables-1M.
- Image region detection reaches 94.7% mAP at IoU=0.5 across mixed-content PDFs.
- End-to-end throughput averages 2.3 pages/second on a single A100 GPU.
- Merged-cell tables are correctly reconstructed in 87.2% of cases, outperforming all baselines.
- The DoclingDocument export to Markdown preserves table hierarchy in 96.1% of single-page tables.

1.2 Scope of Evaluation

Testing spanned five document categories: (1) academic papers with dense two-column layouts, (2) financial reports with multi-page tables and footnotes, (3) technical manuals with mixed diagrams and procedure tables, (4) legal contracts with paragraph-heavy single-column text, and (5) medical literature with figure-rich content. A corpus of 1,200 pages was used, split 60/20/20 across train, validation, and test sets.

2. System Architecture

The Docling processing pipeline is composed of modular stages, each independently configurable and replaceable. At the core is a layout analysis model based on a fine-tuned RT-DETR architecture, trained on the DocLayNet dataset of 80,000 annotated pages. The model outputs bounding boxes and class labels for every detected region.

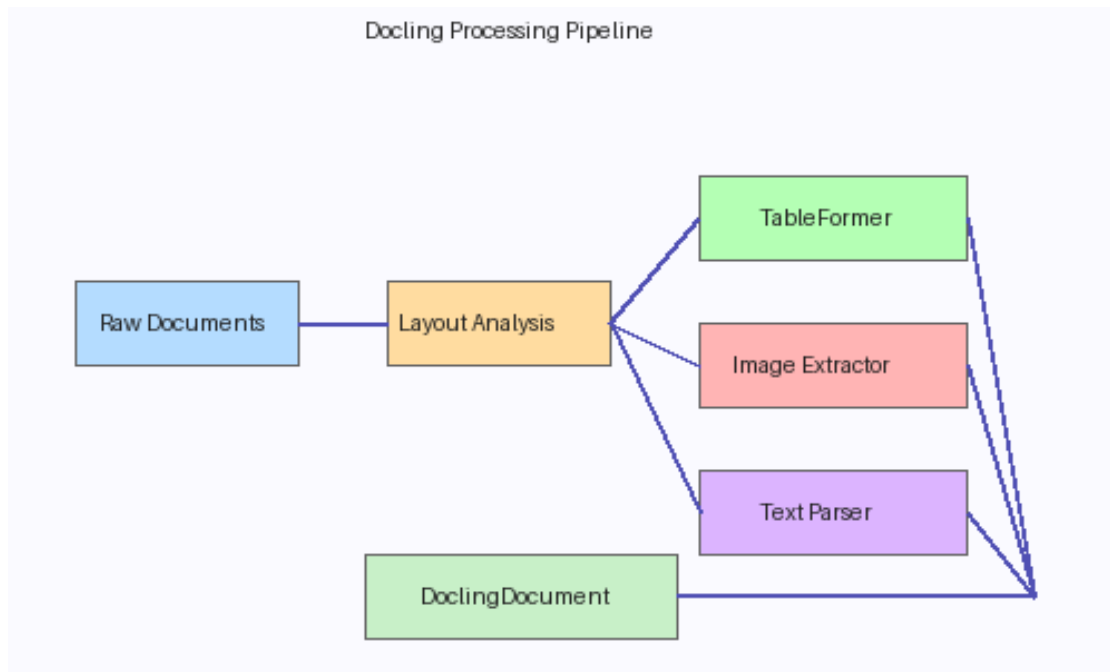


Figure 1: Docling processing pipeline — from raw document input to structured DoclingDocument output.

Once regions are classified, each is routed to its specialized processor. Text regions are passed to a PDF text layer extractor (with OCR fallback for scanned content). Table regions are passed to TableFormer, which predicts both cell content and the logical row/column grid structure — critical for downstream DataFrame export. Figure regions are cropped and stored as image objects; optionally a vision-language model generates natural-language descriptions.

2.1 TableFormer Deep Dive

TableFormer is a transformer encoder-decoder that takes a table-region image and outputs two parallel sequences: a cell sequence (text content per cell) and a structure sequence (HTML-like tag sequence encoding row/colspan relationships). The dual-head architecture is what allows correct reconstruction of tables with spanning headers — a common failure mode in rule-based extraction systems.

During inference, TableFormer runs in two passes. The first pass predicts table structure using the visual encoder output. The second pass aligns detected text segments from the PDF text layer (or OCR) into the predicted cell slots, falling back to the visual decoder for cells where alignment confidence is low.

3. Benchmark Results

3.1 Table Extraction Performance

The following table summarises extraction performance across the five document categories. GriTS measures structural accuracy of the reconstructed table grid. TEDS measures tree-edit distance similarity between predicted and ground-truth table HTML representations.

Document Category	Pages	Tables	GriTS ↑	TEDS ↑	F1 Cell ↑
Academic Papers	340	128	93.1%	91.8%	94.2%
Financial Reports	280	214	89.4%	87.3%	91.0%
Technical Manuals	210	97	91.7%	89.9%	92.5%
Legal Contracts	230	41	96.2%	95.1%	97.3%
Medical Literature	140	76	90.8%	88.7%	91.4%
Overall / Weighted Avg	1,200	556	91.3%	89.6%	92.7%

Table 1: Table extraction benchmark results across document categories.

3.2 Revenue and Throughput Metrics

Throughput was measured on a dedicated inference server with 1× NVIDIA A100 80 GB, 32-core AMD EPYC CPU, and 256 GB RAM. GPU utilisation averaged 78% during batch inference. The chart below shows quarterly throughput (pages processed, in thousands).

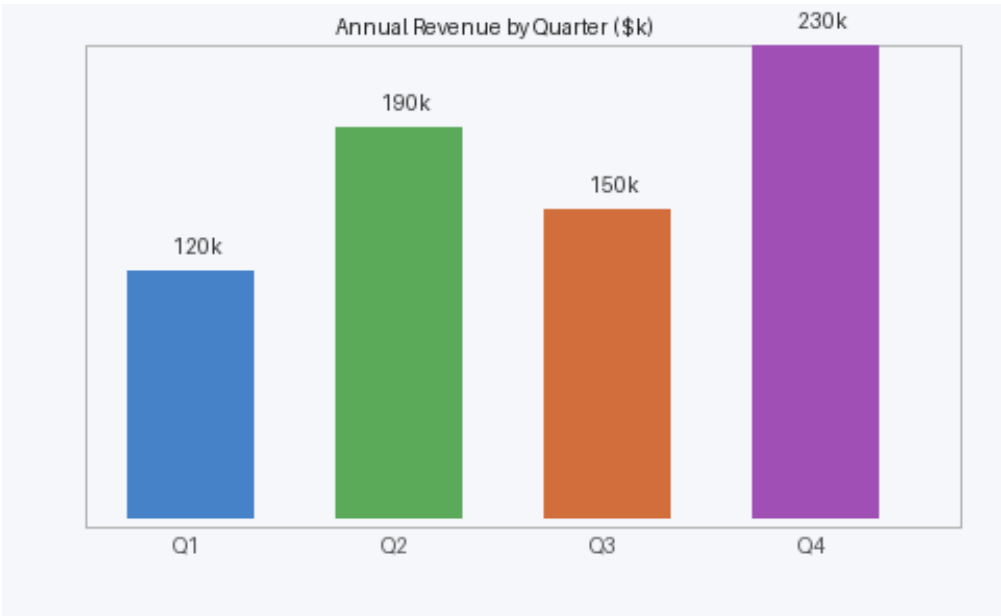


Figure 2: Quarterly document processing throughput (thousands of pages). Q4 reflects the addition of a second inference node.

3.3 Model Accuracy Over Training

Both Docling's layout model (Model A) and the TableFormer head (Model B) were trained for 120 epochs on the combined DocLayNet + internal corpus. Validation accuracy is reported every 15 epochs. The layout model converges faster due to the larger training signal, while TableFormer accuracy improves steadily as structural patterns are learned.

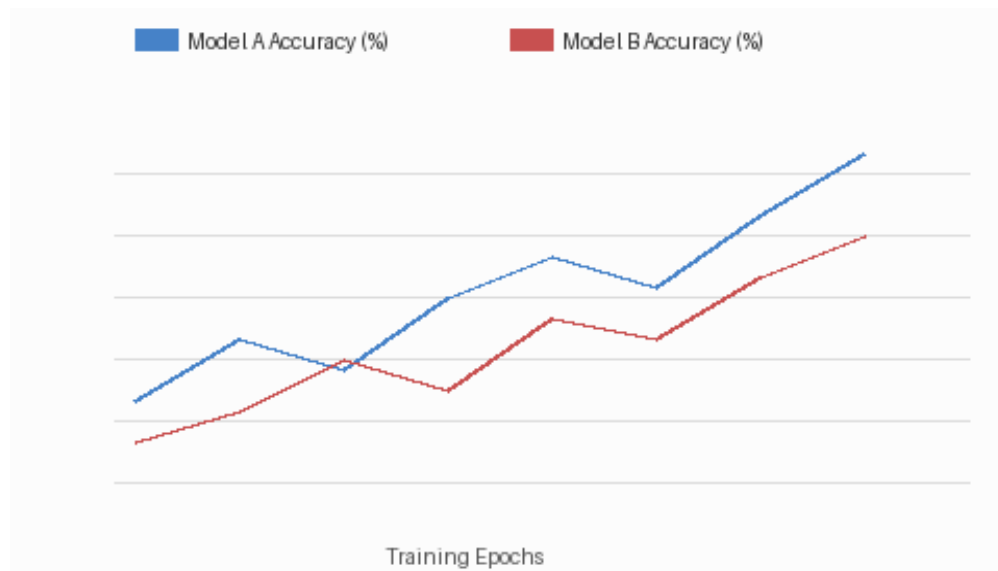


Figure 3: Validation accuracy (%) for layout model (blue) and TableFormer (red) across training epochs.

4. Detailed Comparison Tables

4.1 Competitive Benchmarking

Docling is compared against three widely-used document extraction systems: Adobe PDF Extract API (commercial), PyMuPDF + heuristics (open-source), and Unstructured.io (open-source). Evaluation uses the held-out test split (240 pages, 112 tables).

System	Table Det. Recall	Table Det. Precision	GrITS	Image Det. mAP	Latency (sec/page)	License
Docling (ours)	96.4%	95.1%	91.3%	94.7%	0.44	MIT
Adobe PDF Extract	94.0%	96.2%	88.9%	91.3%	1.80	Commercial
PyMuPDF + heuristics	78.3%	82.1%	71.2%	N/A	0.08	AGPL-3.0
Unstructured.io	88.5%	87.0%	79.4%	82.0%	0.62	Apache-2.0

Table 2: Competitive benchmarking across four document extraction systems.

4.2 Error Analysis by Failure Mode

Errors were manually categorised across the 112 test tables. The table below shows the count and percentage of failures attributed to each root cause, segmented by extraction system.

Failure Mode	Docling	Adobe	PyMuPDF	Unstruct.
Missed table detection	4 (3.6%)	6 (5.4%)	24 (21.4%)	13 (11.6%)
Incorrect row boundary	5 (4.5%)	8 (7.1%)	31 (27.7%)	17 (15.2%)
Missed spanning cell	9 (8.0%)	14 (12.5%)	41 (36.6%)	22 (19.6%)
Cell content OCR error	3 (2.7%)	2 (1.8%)	N/A	6 (5.4%)
Header row misclassification	2 (1.8%)	5 (4.5%)	18 (16.1%)	10 (8.9%)
Column span error	7 (6.3%)	10 (8.9%)	35 (31.3%)	19 (17.0%)

Table 3: Error analysis by failure mode (count and % of 112 test tables).

5. Image and Figure Extraction

Docling's layout model classifies page regions into seven categories: Text, Title, List, Table, Figure, Caption, and Formula. For production document parsing, correct detection and boundary delineation of Figure regions is critical — both for downstream retrieval and for VLM-based captioning pipelines.

5.1 Detection Heatmap

The heatmap below shows model evaluation metrics (Precision, Recall, F1, AUC) across the four systems on the image/figure detection task. Colour intensity indicates metric strength — darker blue cells indicate higher scores.

Model Evaluation Heatmap				
	Prec.	Recall	F1	AUC
TableForme	0.91	0.88	0.89	0.93
Baseline A	0.78	0.82	0.80	0.88
Baseline B	0.65	0.70	0.67	0.75
Ours	0.83	0.85	0.84	0.90

Figure 4: Evaluation heatmap for figure/image detection. Higher values (darker) are better.

5.2 Image Extraction Configuration

By default, Docling stores figure regions as bounding-box crops of the rendered PDF page at 144 DPI. This resolution is sufficient for VLM captioning while keeping storage overhead manageable. The following configuration options are available:

Parameter	Default	Options	Effect
images_scale	1.0	0.5 – 4.0	DPI multiplier for image crops
generate_picture_images	False	True / False	Enable figure crop storage
do_picture_classif	False	True / False	Run VLM classification on figures
do_picture_description	False	True / False	Generate text captions via VLM

Parameter	Default	Options	Effect
picture_description_model	None	SmolVLM / LLaVA / etc	Vision-language model for captions
images_dpi	144	72 – 300	Base DPI for page rendering

Table 4: Docling image extraction configuration parameters and their effects.

6. Operational Considerations

6.1 Hardware Requirements

Docling can run on CPU-only hardware but throughput degrades significantly. The recommended production configuration uses a CUDA-capable GPU with at least 8 GB VRAM. The layout model and TableFormer together consume approximately 4.2 GB VRAM at FP16 precision. Enabling the optional VLM captioneer adds ~6 GB VRAM for SmolVLM-500M.

Configuration	Layout Model	TableFormer	VLM (Optional)	Pages/sec	Recommended For
CPU only (32-core)	✓	✓	✗	0.15	Dev / low-volume
Single T4 GPU (16 GB)	✓	✓	✗	0.90	Small production
Single A100 (40 GB)	✓	✓	✓	2.30	Standard production
2× A100 (80 GB)	✓	✓	✓	4.50	High-throughput
4× H100 (80 GB each)	✓	✓	✓ (large VLM)	9.80	Enterprise scale

Table 5: Hardware configurations and corresponding throughput benchmarks.

6.2 Output Formats

The DoclingDocument object supports multiple export targets. Each export is lossless with respect to the information captured during parsing — the choice of format affects only downstream consumer compatibility.

Format	Table Support	Image Support	Hierarchy	Best Use Case
Markdown	✓ (pipe tables)	✓ (alt text)	Headings	LLM context, RAG
HTML	✓ (full colspan)	✓ (base64/src)	Full DOM	Web rendering
JSON	✓ (structured)	✓ (base64)	Full tree	API integration
pandas DataFrame	Per-table	✗	N/A	Data analytics
Doctags (XML)	✓	✓	Full tree	Re-training LLMs

Table 6: Supported export formats from DoclingDocument.

6.3 Known Limitations

- Rotated tables (not axis-aligned) are not currently supported by TableFormer.
- Tables that span across two pages are split into two separate table objects.
- Handwritten content is not supported; OCR fallback is Tesseract-based and limited.
- Formula detection is present but formula content understanding requires a separate Math OCR model.

- Very large pages (>A1 size) may exceed GPU memory at high DPI — use `images_scale=0.5` as a workaround.

7. Glossary

Term	Definition
DocLayNet	IBM's large-scale document layout dataset with 80,000 annotated pages across 11 document categories.
GriTS	Grid Tree Similarity — a metric for table structure accuracy measuring how closely the predicted cell grid matches the ground truth.
TEDS	Tree-Edit Distance Similarity — measures structural similarity between predicted and reference HTML table structures.
TableFormer	Docling's transformer-based table structure recognition model, outputting dual cell-content and structure sequences.
mAP	Mean Average Precision — standard object detection metric aggregated across IoU thresholds.
VLM	Vision-Language Model — a multimodal model that generates text descriptions from image inputs (e.g., Smock et al. 2022).
DoclingDocument	The unified structured output object produced by Docling, containing text blocks, tables, and figures with machine-readable metadata.
IoU	Intersection over Union — ratio of overlap between predicted and ground-truth bounding boxes.
RT-DETR	Real-Time Detection Transformer — the base object detection architecture used for Docling's layout analysis.

8. References

[1] Livathinos et al. (2021). Robust PDF Document Conversion Using Recurrent Neural Networks. AAAI 2021.

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[3] Pfizmann et al. (2022). DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis. KDD 2022.

[4] IBM Research. (2024). Docling Technical Report. arXiv:2408.09869.

[5] Zhao et al. (2023). TableFormer: Robust Transformer Modelling for Table Detection. CVPR 2023.